

git 사용법



버전 관리



최종.ppt



최종_수정.ppt



진짜최종.ppt



진짜최최종.ppt



진짜최최최
종.ppt



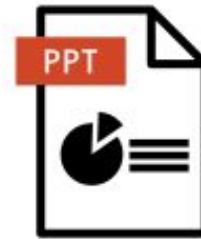
진짜최최최종
_V2.ppt



파이널최종.ppt



진짜파이널
최종.ppt



진짜파이널
최최종.ppt

SCM(Source Code Management)

SCM

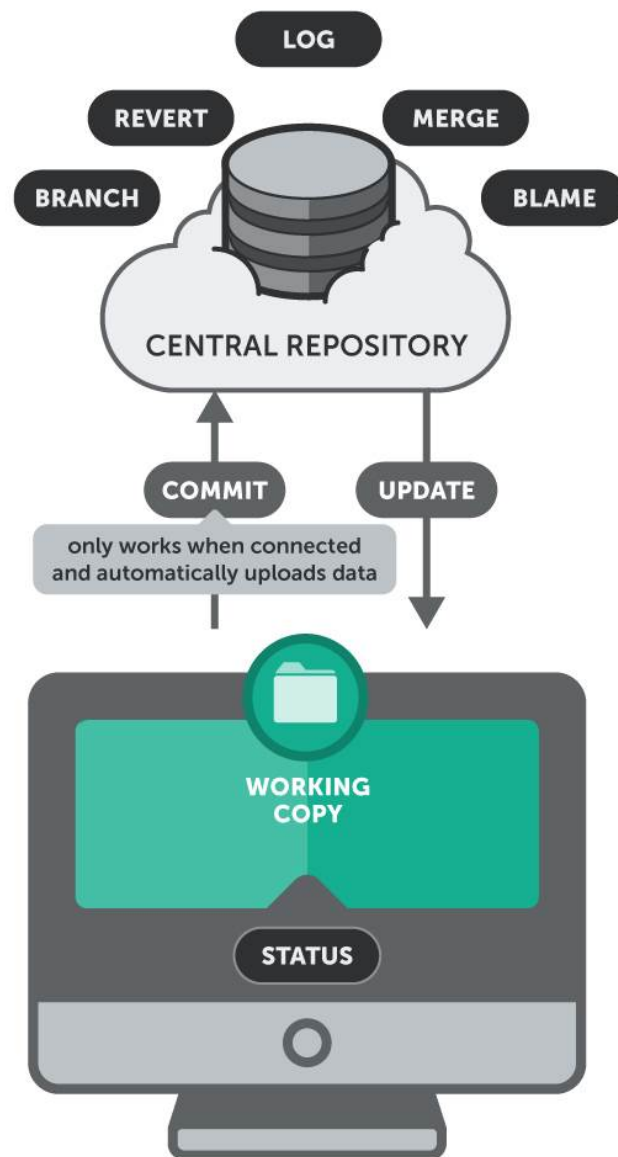
(Source Code Management)



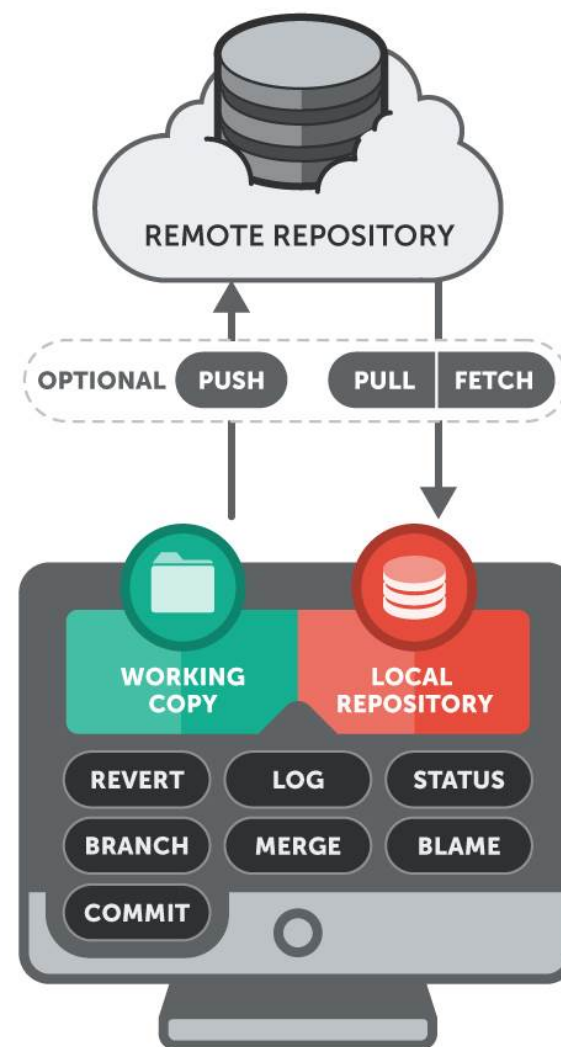
TOP VERSION CONTROL SYSTEMS

SVN vs Git

SUBVERSION



GIT



git 설치

 **git** --fast-version-control

Type / to search entire site...

Git is a [free and open source](#) distributed version control system designed to handle everything from small to very large projects with speed and efficiency.

Git is [lightning fast](#) and has a huge ecosystem of [GUIs](#), [hosting services](#), and [command-line tools](#).



**About**
Git's performance and ecosystem

**Learn**
Pro Git book, videos, tutorials, and cheat sheet

**Tools**
Command line tools, GUIs, and hosting services

**Reference**
Git's reference documentation

**Install**
Binary releases for all major platforms.

**Community**
Get involved! Bug reporting, mailing list, chat, development and more.



Latest source release
2.53.0
[Release Notes](#) (2026-02-02)

Install for Mac

GitHub Repository

git-scm.com

환경설정

사용자 이름 설정

```
git config --global user.name "사용자 이름"
```

사용자 이메일 설정

```
git config --global user.email "이메일 주소"
```

맥사용시 파일명이 한글일경우 인식문제 해결

```
git config --global core.precomposeunicode true
```

git에서 한글 깨짐이 발생하는 경우

```
git config --global core.quoteopath false
```

config 내용 확인

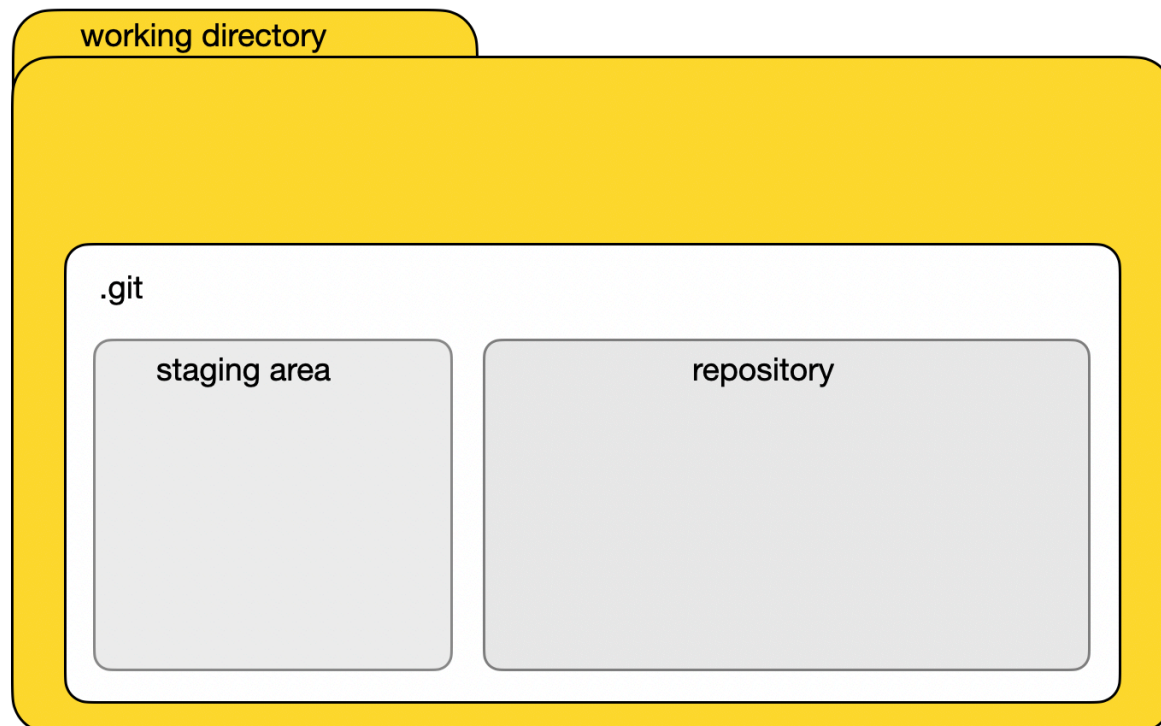
```
git config --list
```


저장소 생성

현재 폴더를 Git 로컬 저장소로 생성

git init

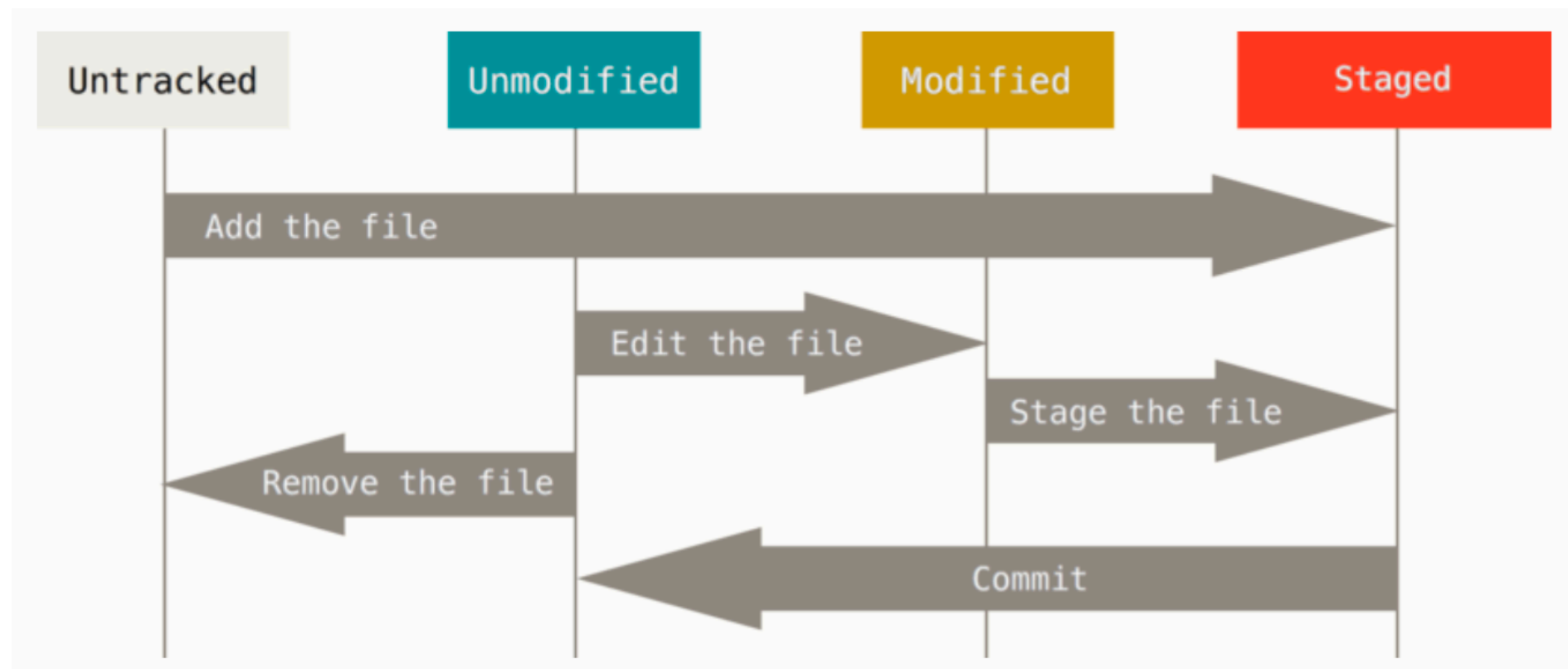
```
→ git-test git init
Initialized empty Git repository in /Users/wizard/git-test/.git/
→ git-test git:(master) ls -al
total 0
drwxr-xr-x   3 wizard  staff   96 Jan  1 21:52 .
drwxr-x---+ 106 wizard  staff 3392 Jan  1 21:52 ..
drwxr-xr-x   9 wizard  staff  288 Jan  1 21:52 .git
```



파일의 상태

현재 저장소의 상태 확인

git status



저장소 생성 실습

- vscode에서 폴더를 생성
- a.txt 만들고 a 라고 내용 작성
- b.txt 만들고 b 라고 내용 작성
- 터미널에서 git init 실행
- 터미널에서 git status 실행

```
On branch master
```

```
No commits yet
```

```
Untracked files:
```

```
(use "git add <file>..." to include in what will be committed)
```

```
a.txt
```

```
b.txt
```

```
nothing added to commit but untracked files present (use "git add" to track)
```

파일등록

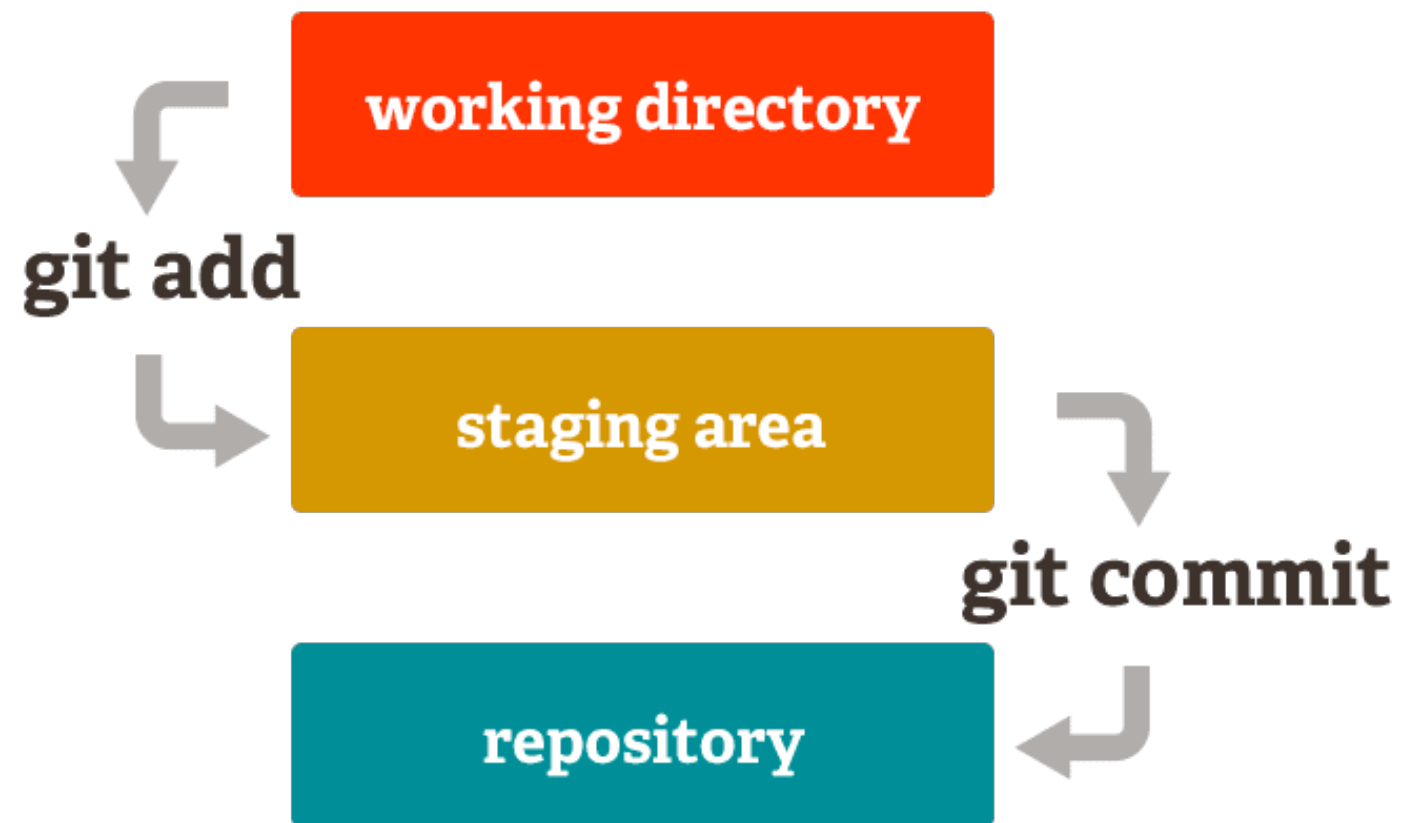
특정 파일을 staging area에 추가

`git add "파일명"`

현재 폴더의 모든 파일 추가

`git add -A`

`git add .`



파일등록 실습

- 터미널에서 `git add .` 실행
- 터미널에서 `git status` 실행

```
On branch master
```

```
No commits yet
```

```
Changes to be committed:
```

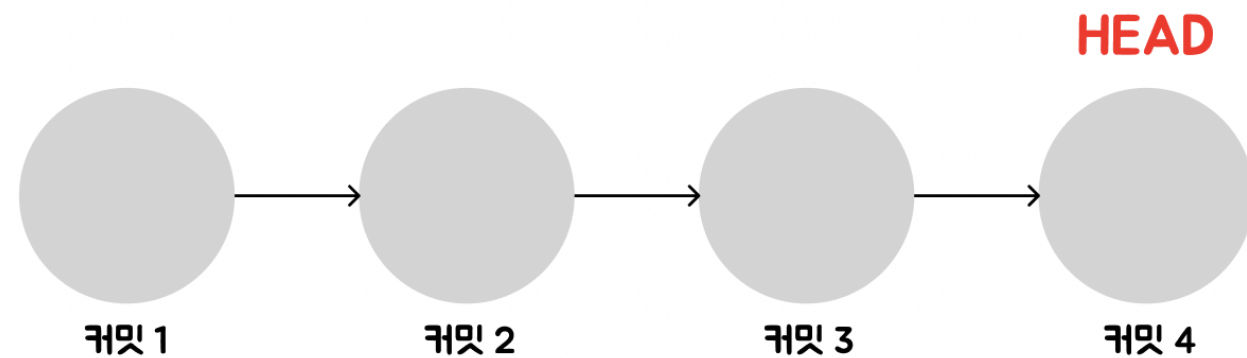
```
(use "git rm --cached <file>..." to unstage)
```

```
new file:   a.txt
```

```
new file:   b.txt
```

커밋(Commit)

- 저장소의 변경 내용을 특정 시점에 저장하는 행위
- 저장된 덩어리



커밋 실습

현재 폴더의 변경사항을 확인

```
git status
```

현재 상태에서 커밋 생성

```
git commit -m "commit message"
```

현재 저장소의 커밋 내역을 확인

```
git log
```

```
git log --oneline
```

commit 실습

- 터미널에서 `git commit -m "init commit"`

```
[master (root-commit) 4cfaa19] init commit
2 files changed, 2 insertions(+)
create mode 100644 a.txt
create mode 100644 b.txt
```

- 터미널에서 `git log` 실행

```
commit 4cfaa192aeb9d24e07022e68d467e5bfb1756e0d (HEAD -> master)
Author: wizard113 <yeong.rok.lee@gmail.com>
Date: Tue Nov 29 21:57:54 2022 +0900

    init commit
```


commit 실습

- c.txt 추가하고 파일 내용으로 c를 작성
- 터미널에서 git status 실행

```
On branch master
```

```
Untracked files:
```

```
(use "git add <file>..." to include in what will be committed)
```

```
c.txt
```

```
nothing added to commit but untracked files present (use "git add" to track)
```

- git add . 실행
- git commit -m "add c.txt" 실행
- git log --oneline 실행

```
[master 8e2232f] add c.txt
```

```
1 file changed, 1 insertion(+)
```

```
create mode 100644 c.txt
```

.

```
8e2232f (HEAD -> master) add c.txt
```

```
4cfaa19 init commit
```

—

commit간 이동

git checkout 6ef2a23 (commit id 앞7자리)

git checkout - (최신 커밋 checkout)

commit 되돌리기

해당 id의 커밋으로 되돌림 해당 커밋이후의 커밋은 삭제됨

```
git reset --hard 6ef2a23
```

해당 id의 커밋으로 새로운 커밋을 만들어 최신 커밋으로 만듦

```
git revert 6ef2a23
```

git reset 실습

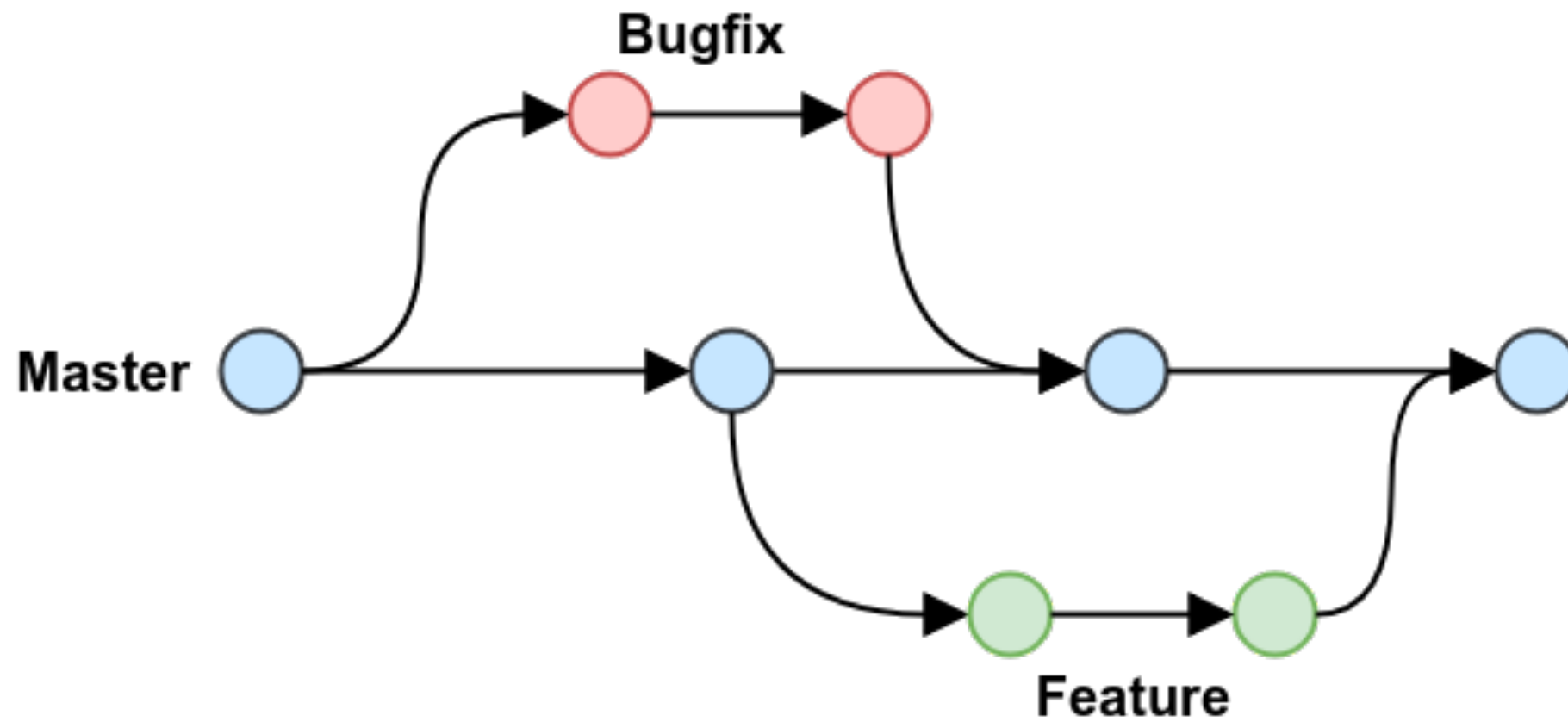
- a.txt에 내용 "git" 추가
- d.txt 추가하고 내용 입력
- git status 실행
- git add . 실행
- git commit -m "세번째 커밋"
- git reset --hard 8e3332f
- 세번째 커밋으로 리셋

git revert 실습

- a.txt에 내용 "git" 추가
- d.txt 추가하고 내용 입력
- git status 실행
- git add . 실행
- git commit -m "세번째 커밋"
- git revert 8e3332f
- 세번째 커밋으로 리셋

```
363f2fe (HEAD -> master) Revert "add c.txt"
148ff65 세 번째 커밋
8e2232f add c.txt
4cfaa19 init commit
```

Branch



Branch

새로운 branch 생성

```
git branch branch_name
```

작업 branch 변경

```
git checkout branch_name
```

```
git switch branch_name
```

새로운 branch를 생성하고 변경

```
git switch -c branch_name
```


branch 실습

- git branch develop
- git branch 실행해서 branch 목록확인

```
develop
* main
```

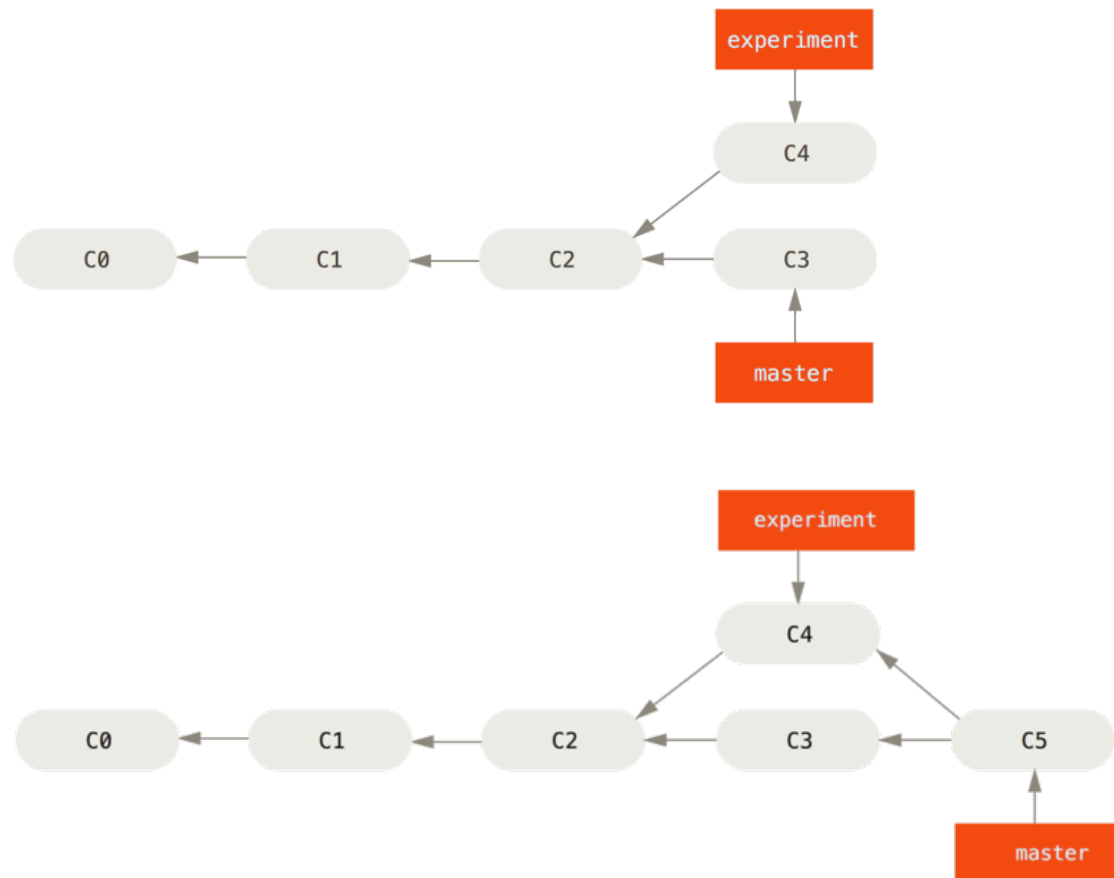
- git switch develop
- git branch

```
* develop
main
```


merge

현재 branch에 특정 branch merge

`git merge branch_name`

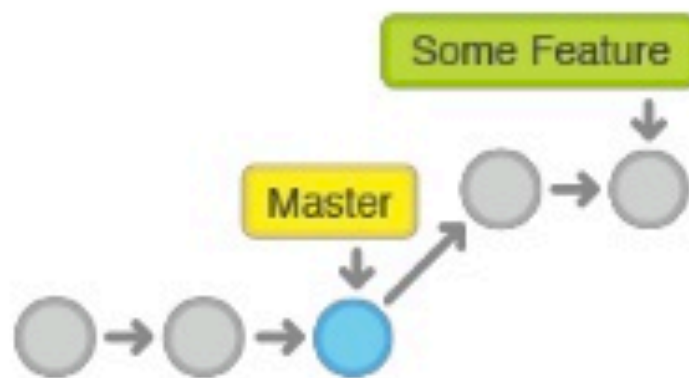


일반적인 commit merge

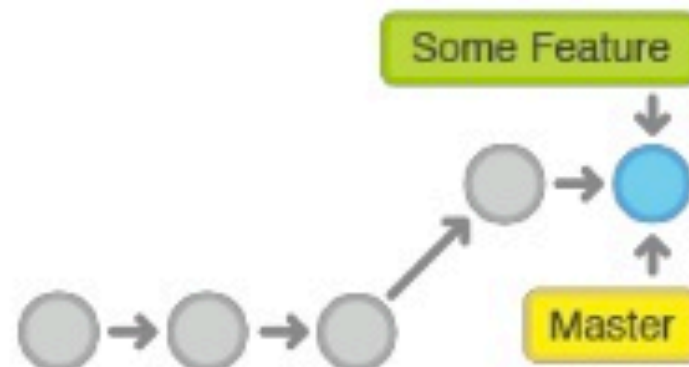
Fast-Forward Merge Illustrated

fast-forward merge

Before Merging



After a Fast-Forward Merge



no-ff merge

Before Merging



After a 3-way Merge

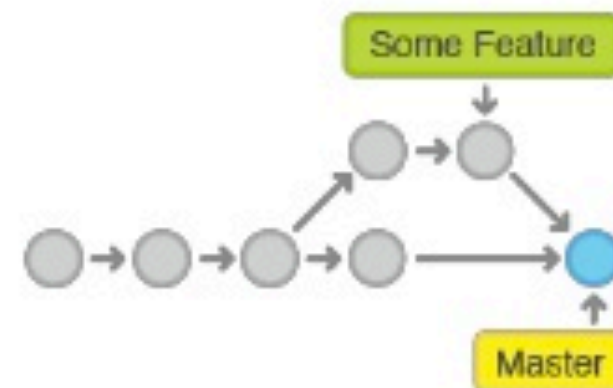
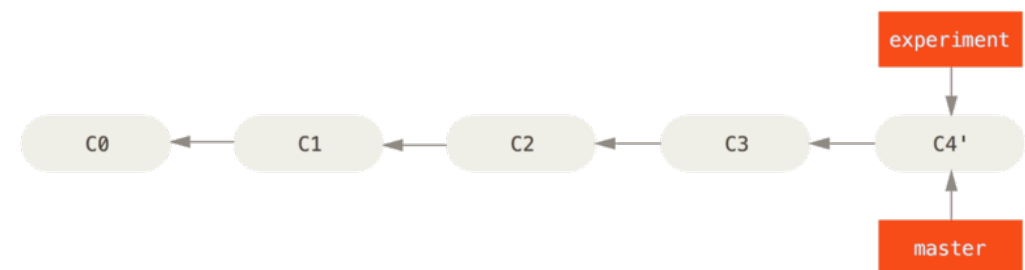
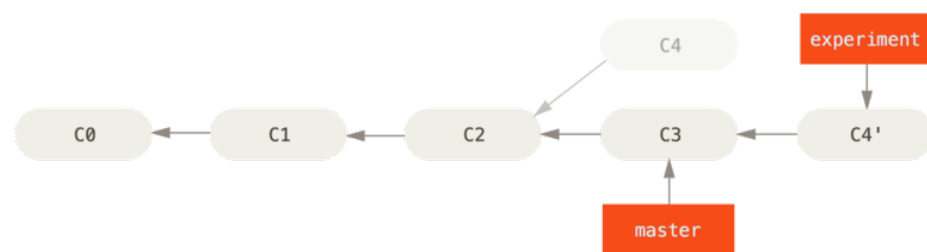
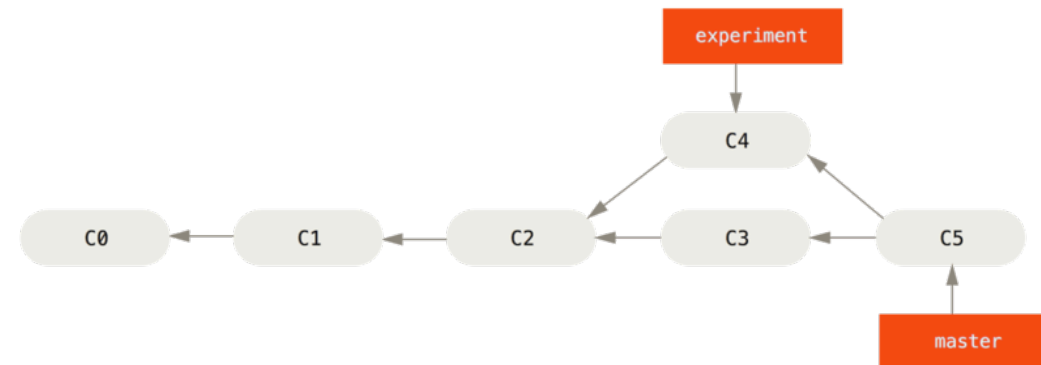
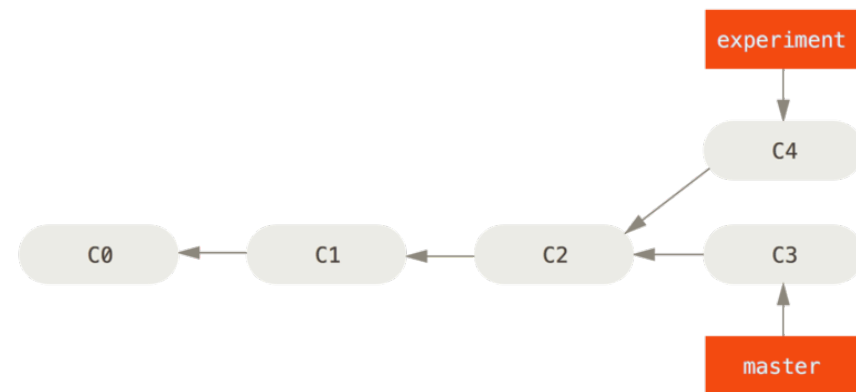


Image: <https://www.atlassian.com/git/tutorial/git-branches#!merge>

rebase

현재 branch에 특정 branch merge

git rebase branch_name



merge 실습

- git branch develop
- a.txt에 "develop branch에서 작업" 내용 추가
- git status
- git add .
- git commit -m "5번째 커밋 on develop"
- git branch main
- git merge develop
- git branch -d develop

Updating 148ff65..6cee7f8

Fast-forward

a.txt | 3 ++-

1 file changed, 2 insertions(+), 1 deletion(-)

merge시 충돌 처리

- git branch develop
- a.txt에 "develop branch에서 작업 1" 내용 추가
- git status
- git add .
- git commit -m "6번째 커밋 on develop"
- git branch master
- a.txt에 "main branch에서 작업 1" 내용 추가
- git status
- git add .
- git commit -m "여섯번째 커밋 on main"

merge시 충돌 처리

- git branch main
- git merge develop
- git branch -d develop

<<<<<<< HEAD (현재 변경 사항)

4 main branch에서 작업

=====

4 develop branch에서 작업1

>>>>>>> develop (수신 변경 사항)

- a.txt를 각 branch에서 동시에 수정해서 충돌 발생
- a.txt를 수정 후 커밋하면 충돌 해소

```
* 47a1206 (HEAD -> main) 7th commit merge develop to main
|\
| * 020d2d9 (develop) 6th commit on develop
* | cb6160c 6th commit on main
|/
* 6f0d3c0 (master) 5th commit on develop
* 922e957 Revert "add c.txt"
* 24e146c 세 번째 커밋
* c453a3a add c.txt
* e780c2e fist commit
```

원격저장소 생성

Create a new repository

A repository contains all project files, including the revision history. Already have a project repository elsewhere? [Import a repository.](#)

Owner *

Repository name *



Great repository names are short and memorable. Need inspiration? How about [studious-invention?](#)

Description (optional)



Public

Anyone on the internet can see this repository. You choose who can commit.



Private

You choose who can see and commit to this repository.

Initialize this repository with:

Skip this step if you're importing an existing repository.



Add a README file

This is where you can write a long description for your project. [Learn more.](#)

Add .gitignore

Choose which files not to track from a list of templates. [Learn more.](#)

.gitignore template: None ▾

Choose a license

A license tells others what they can and can't do with your code. [Learn more.](#)

License: None ▾

You are creating a public repository in your personal account.

Create repository

원격저장소 생성

Quick setup — if you've done this kind of thing before

 Set up in Desktop

or

HTTPS

SSH

<https://github.com/jellyworks/nambu2022.git>



Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#).

...or create a new repository on the command line

```
echo "# nambu2022" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/jellyworks/nambu2022.git
git push -u origin main
```



...or push an existing repository from the command line

```
git remote add origin https://github.com/jellyworks/nambu2022.git
git branch -M main
git push -u origin main
```



...or import code from another repository

You can initialize this repository with code from a Subversion, Mercurial, or TFS project.

[Import code](#)

원격저장소 실습

새로운 원격저장소를 origin이름으로 추가

```
git remote add origin https://github.com... .git
```

master branch명을 main으로 변경

```
git branch -M main
```

origin 원격저장소의 main branch에 push

```
git push origin main
```

원격저장소와 local의 상태를 일치시킴

```
git pull
```

원격저장소 실습

github에 있는 저장소를 clone(main branch)

git clone git주소

github에 있는 저장소를 clone(원하는 branch)

git clone -b branch_name git주소

리모트에 있는 branch도 보여줌

git branch -a

```
main
remotes/origin/HEAD -> origin/main
remotes/origin/dev
remotes/origin/main
```

Pull Request 생성

Open a pull request

Create a new pull request by comparing changes across two branches. If you need to, you can also [compare across forks](#).

 base: main ← compare: dev ✓ **Able to merge.** These branches can be automatically merged.



Dev

Write

Preview

Leave a comment

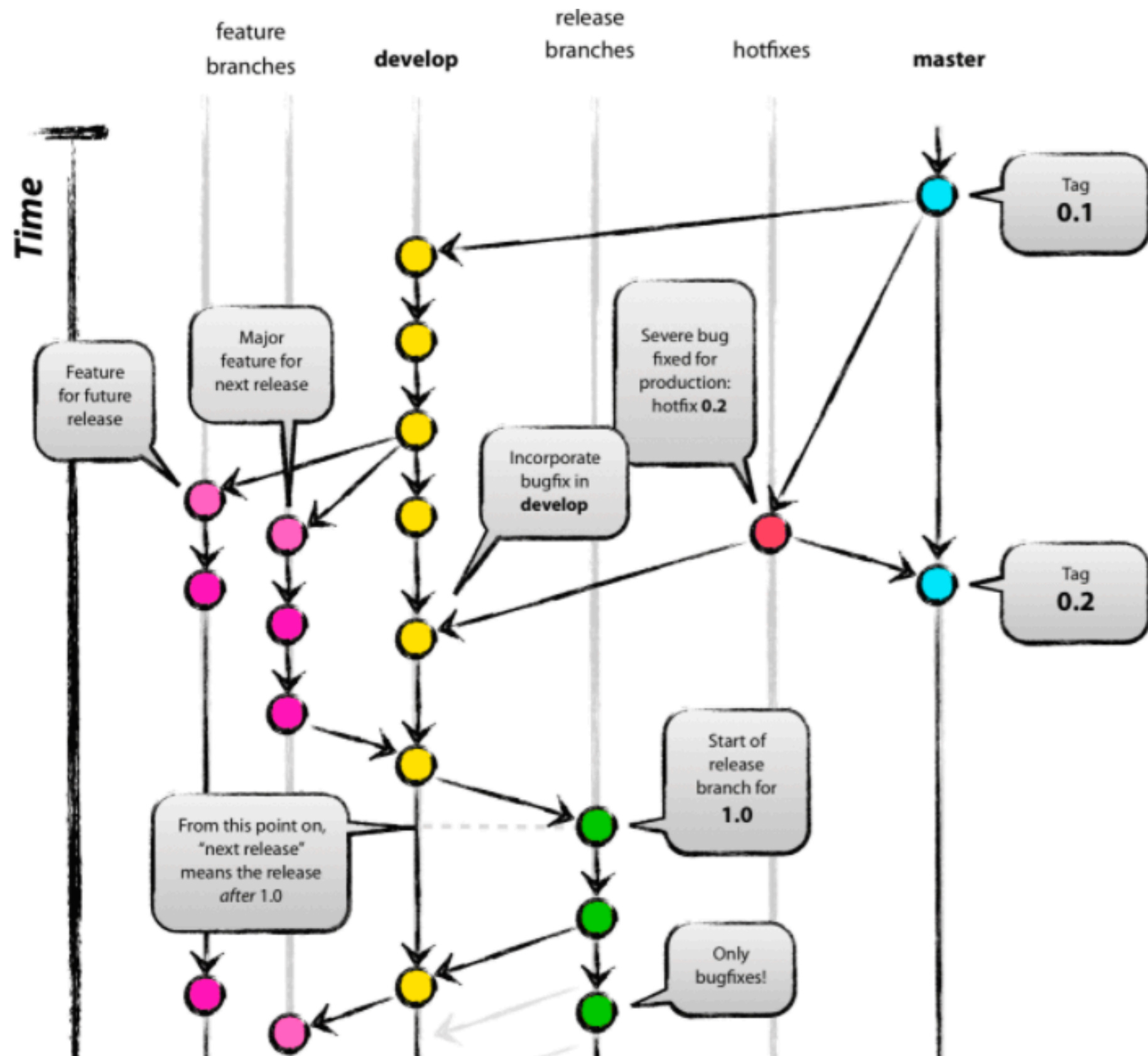
Attach files by dragging & dropping, selecting or pasting them.

Create pull request

▼

 Remember, contributions to this repository should follow our [GitHub Community Guidelines](#).

Git-flow branch 전략



Source Tree

The screenshot shows the Source Tree application interface. The top bar includes icons for commit, pull, push, fetch, branch, merge, stash, log, diff, and settings. The left sidebar contains navigation options: workspace, file status, history (selected), search, branches, tags, log, origin, feature, main, test, local-branch, remote-branch, commit, submodules, and subtreelists.

The main area displays a commit history graph and a list of commits. The graph shows a sequence of commits with a color-coded branch structure. The commit list includes the following details:

커밋	작성자	날짜
ed06180	wizard113 <yeon...	2022. 2. 27. 16:57
2566bc9	wizard113 <yeon...	2022. 2. 27. 16:50
d6fb1d7	wizard113 <yeon...	2022. 2. 27. 16:49
2646905	wizard113 <yeon...	2022. 2. 27. 13:03
13fa32a	azjellyworks <10...	2022. 2. 27. 12:55
b49b59d	azjellyworks <az...	2022. 2. 27. 12:46
f4eb72d	wizard113 <yeon...	2022. 2. 27. 13:01
6750208	Youngrok Lee <y...	2022. 2. 27. 00:31
2b7bd9f	wizard113 <yeon...	2022. 2. 27. 00:...
1b88c90	wizard113 <yeon...	2022. 2. 27. 00:...
518a477	wizard113 <yeon...	2022. 2. 27. 00:...
d2c3d41	wizard113 <yeon...	2022. 2. 26. 23:...
02bc99e	wizard113 <yeon...	2022. 2. 26. 23:...
fb6a1cf	wizard113 <yeon...	2022. 2. 26. 23:...
d867eba	wizard113 <yeon...	2022. 2. 26. 21:...
3b69486	wizard113 <yeon...	2022. 2. 26. 21:...
118f02b	wizard113 <yeon...	2022. 2. 26. 20:11

The bottom section shows a diff view for the file `.github/workflows/ci.yml`. The diff highlights changes in the `node-version` field, showing a transition from `[10.x]` to `[16.x]`.

Commit details for the selected commit (HEAD main origin/main) are as follows:

- 제출 ID: f72f7cb0c0d230328fe65e3d79bc4d861245c43c [f...]
- 상위 항목: 8673f079fb
- 작성자: wizard113 <yeon.rok.lee@gmail.com>
- 날짜: 2022년 2월 27일 22시 54분 16초 GMT+9
- 레이블: HEAD main origin/main